EDUGENIE: GOOGLE GEMINI POWERED LEARNING ASSISTANT

Phase 3: Project Design Phase

The Project Design Phase focused on designing the architecture, modules, APIs, and user interface of EduGenie.

Folder Architecture

```
EduGenie/
│
├── main.py
├── explanation_module.py
├── qna.py
├── quiz_module.py
├── summary_module.py
├── learning_path.py
│
├── templates/
│   └── index.html
│
├── static/
│   └── style.css
│
└── requirements.txt
```

File Description

File	Function
`main.py`	FastAPI application
`explanation_module.py`	Concept explanation

File	Function
`qna.py`	Question answering
`quiz_module.py`	Quiz generation
`summary_module.py`	Summarization
`learning_path.py`	Learning recommendations
`index.html`	Web interface
`style.css`	Frontend styling
`requirements.txt`	Project dependencies

System Design

The basic workflow of the system was designed as:

```
User
 ↓
Web Interface
 ↓
FastAPI Backend
 ↓
Selected Module
 ↓
AI Model
 ↓
Generated Response
 ↓
Web Interface
```

The modular architecture allows each functionality to be developed and maintained independently.